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Comparison between visual inspection, dermoscopy, and reflectance confocal microscopy in the diagnosis of lentigo maligna – a systematic review

Lentigo maligna (LM) and lentigo maligna melanoma (LMM) are melanoma subtypes associated with chronic photoexposure, often located on areas such as the face and neck. Early diagnosis of these lesions is challenging due to their similarity to other benign conditions. Non-invasive methods, such as visual inspection, dermoscopy (DMT), and reflectance confocal microscopy (RCM), have been employed to improve diagnostic accuracy and surgical margin definition. However, evidence on which technique is superior remains uncertain. This systematic review aimed to investigate whether DMT is more effective than other non-invasive methods, such as RCM and visual inspection, for the diagnosis of LM and LMM confirmed by histopathology. The literature research was conducted in PubMed, CINAHL, MEDLINE, and EMBASE databases without language and date of publication restrictions. Two independent dermatologists assessed the eligibility of the studies, extracted data, and performed a methodological quality and risk of bias analysis of the included studies (QUADAS-2). From 532 retrieved studies, 11 were included in the final analysis. The included studies showed substantial heterogeneity in design, thresholds, outcomes, and reporting, which precluded quantitative analysis. Compared with histopathology, the diagnostic performance of non-invasive methods, such as visual inspection, DMT, and RCM, was inconclusive due to several limitations. For instance, we found strong evidence of selection biases, non-consecutive sampling, lack of randomization, unblinded evaluators, and under-reporting of sensitivity and specificity. Overall, current evidence does not support replacing histopathology for confirming LM and LMM diagnosis. Standardized study protocols and reporting in the field of non-invasive diagnosis of LM and LMM are needed.

Key words: lentigo maligna, melanoma, dermoscopy, systematic review, skin cancer

European countries exhibit the highest incidence of UV-attributable skin cancer worldwide [1], with increasing rates in several countries [2]. Melanoma is the most severe neoplasm, as it is associated with a higher probability of causing metastases, including to the lung and brain [3]. Among all skin cancers, lentigo maligna (LM) and lentigo maligna melanoma (LMM) are predominantly linked to chronic photoexposure in areas such as the face and neck [4, 5]. In Europe, LM and LMM have shown an increase in incidence in recent years [6, 7] evidencing an urgent need for identifying optimal methods for their diagnosis and treatment. However, both LM and LMM lesions arise in cosmetically and functionally critical sites and may be mistaken for non-malignant lesions resulting from chronic photoexposure [8]. Thus, the diagnosis and differentiation of LM and LMM still represent a clinical challenge [9].

Among the non-invasive methods for LM and LMM diagnosis, visual inspection and dermatoscopy (DMT) are commonly used; however, both methods have limitations regarding the adequate assessment of the clinical and subclinical characteristics of these lesions [10]. In addition, reflectance confocal microscopy (RCM) is also a non-invasive diagnostic method that provides high-resolution images at multiple depths in parallel [11]. Although RCM holds great potential for the diagnosis of malignant skin lesions, it remains a limited resource in developing countries, being available only in major centres [12]. Furthermore, the complexity of the lesion

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and its features pose a challenge to the dermatologist in clinical practice, whose goal is to diagnose and treat skin lesions early to reduce the risks of surgery in sensitive areas and the occurrence of metastases.

Although the results obtained with DMT and RCM for the differential diagnosis of LM and LMM appear promising, the definitive use of each method remains controversial. To date, the majority of published studies on non-invasive methods for the diagnosis of LM and LMM have been conducted with small samples [13] or possess experimental designs that limit the extrapolation of evidence to clinical practice [14, 15]. Current studies on non-invasive skin imaging vary widely in quality and scope, leaving unresolved which technique best distinguishes benign lesions from LM and LMM. To address this gap, we performed a systematic review to determine the diagnostic accuracy of DMT, RCM, and other non-invasive methods for LM and LMM.

Methods

Protocol and registration

The guidelines of the PRISMA-DTA (Preferred Reporting Items for Systematic Review and Meta-Analysis extension for Diagnostic Test Accuracy) were employed throughout this study. The project was registered in the International Prospective Register of Systematic Reviews (PROSPERO) under protocol number CRD42023428422.

Eligibility criteria

Our search was conducted based on the population, index test, reference standard, and diagnosis of interest (PIRD) question, which defines a population of adults with skin lesions, with index test being any type of non-invasive dermatological analysis (DMT, RCM, visual inspection), and reference standard test being histopathology, which confirmed the diagnosis of LM and LMM.

Information sources and search

The literature search was conducted in national and international health research portals and biomedical journal databases (CINAHL, MEDLINE, and EMBASE) without publication date and language restrictions. Studies were selected exclusively when they were published in scientific journals. Searches in databases were carried out between 2023 and 2024. The search descriptors are listed in *supplementary table 1*. All searches were performed with the assistance of an experienced librarian. Contact with authors was made when data was not readily available in published studies or unclear data presentation was identified.

Study selection

Two reviewers (BGF, LF) independently screened titles and abstracts and full texts using our inclusion and exclusion criteria. Disagreements were resolved by discussion between reviewers. Then, whole articles were screened by both reviewers and then selected for data extraction. The following articles were included: (a) studies employing case-control, cohort, cross-sectional or clinical trial designs on diagnostic tools in LM/LMM; (b) studies investigating the role of DMT and other non-invasive diagnostic tools for LM and LMM; and (c) articles published in English, French, Italian, Portuguese, or Spanish. We excluded case reports/series, narrative/systematic reviews, editorials, conference abstracts, and studies lacking an appropriate reference standard or extractable data. Selected studies through first screening that had no clear data on the diagnosis of LM and LMM were also removed. Studies with incorrect designs were also excluded from our analyses.

Data collection process

Initially, year of publication, authorship, and study design (clinical trial, case-control, or cohort) were independently extracted from the selected articles by the two reviewers upon full analysis of studies. We also obtained information on outcomes, number of patients investigated, types of lesions examined, index tests, reference standards, presence of randomization of samples, description if evaluators were blinded to the analysis, and whether the diagnosis was confirmed by histopathology before or after lesions were analysed. Subsequently, data on the sensitivity and specificity of each diagnostic method (visual inspection, DMT, and RCM) were gathered from the selected studies. Results from Receiver Operating Characteristic (ROC) curves and the Area Under the Curve (AUC) were also extracted when available. Descriptive characteristics of each method used for LM and LMM diagnosis were also obtained. Data at all levels of significance and thresholds were gathered. The information from each text was extracted, synthesized, and recorded in protocol forms on the RAYYAN platform to enable the two evaluators to verify validity and applicability of the selected studies.

Definitions for data extraction

We defined the target condition as lentigo maligna (LM) and lentigo maligna melanoma (LMM) when lesions were explicitly identified as such and the diagnosis was confirmed by biopsy or complete excision with histopathology. The index tests included RCM, DMT, and visual inspection as reported by the primary studies. The reference standard was histopathology exclusively; we did not restrict inclusion by follow-up duration or by ongoing/previous treatments. We included studies from any clinical setting available. Units of analysis could be per-patient or per-lesion, with per-patient preferred. We incorporated studies with and without expert blinding to the reference standard.

Risk of bias and applicability

The same two reviewers independently assessed risk of bias and issues about applicability using QUADAS-2 [16]. Disagreements were resolved by consensus between reviewers. For the implementation of the QUADAS-2

protocol, the RevMan (Cochrane) software was used, through which responses were provided for each item, and summary figures were generated.

Diagnostic accuracy measures

We extracted sensitivity, specificity, area under the ROC curve (AUC), ROC curve data (including thresholds when reported), and regression model outputs related to diagnostic performance for all eligible index tests in the included studies. The unit of assessment was per-patient whenever available. Per-lesion data were extracted when per-patient data were not reported, with per-patient being prioritized. Where necessary, we also derived 2×2 data from reported results to ensure consistency across studies.

Synthesis of results

We did not transform, impute, or statistically pool data. Accuracy results (sensitivity, specificity, AUC, ROC information, regression outputs) were extracted exactly as reported; when 2×2 data were reconstructible, we derived them but performed no meta-analysis. To reflect technological and operator evolution, we used a structured narrative synthesis ordered chronologically by years (1980s to 2020s) and grouped the studies by index test (visual inspection DMT, RCM).

Results

Study selection

Across databases, a total of 532 studies were identified. Of these, 203 were retrieved from PubMed, 121 from MEDLINE, 181 from Embase, and 27 from CINAHL. After removing 343 duplicate entries, 189 studies remained for further analysis. These 189 studies underwent abstract screening, during which 117 were excluded. Subsequently, 72 studies were assessed in full by two independent reviewers, resulting in the exclusion of 61 studies and the selection of 11 publications for quality assessment and data extraction. The PRISMA flowchart, summarizing the studies included, screened, and selected, is presented in *figure 1*.

Study characteristics

Eleven studies were selected and analysed in this systematic review, comprising 1,149 patients (*table 1*). The oldest selected study was published in 1983, and the most recent in 2023. The majority of the studies included for analysis were conducted by European research groups, particularly from Italy (54%, 6/11). The sample sizes of the studies ranged from 17 to 318. Of the 11 studies, seven examined both LM and LMM lesions, while four focused exclusively on LM lesions. Diagnostic technique comparisons varied among the studies, but the majority (90%, 10/11) investigated DMT and RCM, either directly or indirectly. Only three studies included comparisons between DMT and RCM with visual inspection [17-19].

The majority of the selected studies investigated both male and female patients, with an average female representation of 52%. The mean age of participants ranged from 65 to 72 years, while the minimum and maximum ages observed across studies were 26 and 97, respectively. Two studies reported a higher proportion of women in their samples [18, 20], with female participation reaching up to 70%.

Most of the included studies investigated LM and LMM lesions, except for Gao *et al.* [18] and Licata *et al.* [19], which focused exclusively on LM lesions. All studies, with the exception of one (Gao *et al.* [18]), evaluated histopathological diagnostic outcomes for LM or LMM. The aforementioned study examined the definition of surgical margins and the number of layers required for Mohs surgery. Other studies also assessed surgical margin definition as an outcome [20-22]. Only one study investigated the presence of disease recurrence following surgical excision of the lesions [20].

Due to the evolution of diagnostic and operator performance for DMT and RCM, we also conducted analyses by decade to account for potential superior results of diagnostic techniques after definition on use standards [23]. Decade grouping did not reveal a consistent temporal trend in diagnostic accuracy. Our set includes one study from the 1980s (visual inspection vs DMT) and no eligible studies from the 1990s or 2000s. Most studies were published in the 2010s (2013-2018) and 2020s (2021-2023). However, heterogeneity in design, outcomes (*e.g.*, margin mapping vs diagnostic accuracy), thresholds, and frequent absence of extractable 2×2 data impaired direct between-decade comparisons. RCM first appears in the 2010s and was consistently evaluated thereafter, while DMT was present from the 1980s onwards. Visual inspection features in early work and in a minority of recent studies, but was not reported across every decade.

Risk of bias and applicability

We applied the QUADAS-2 tool, which evaluates four major domains commonly associated with biased interpretations of results in published research. The QUADAS-2 tool was implemented following data extraction from the articles and provided both overall and domain-specific information regarding the risk of bias in the studies included in this systematic review (*figure 2*). In general, most of the included studies exhibited an intermediate to high risk of bias. However, one study demonstrated a low risk of bias across all domains [24]. We also observed that only five studies had blinded evaluators, three performed direct comparisons of the diagnostic methods, and only one study reported the use of consecutive samples. Lastly, only three studies enrolled patients without prior histopathological confirmation of LM and LMM diagnosis.

Results of individual studies

Nine out of the 11 studies aimed to compare the accuracy of DMT and RCM for the diagnosis of LM and LMM [18-22, 25-28], three aimed to analyse visual inspection versus other diagnostic methods [17-19], and

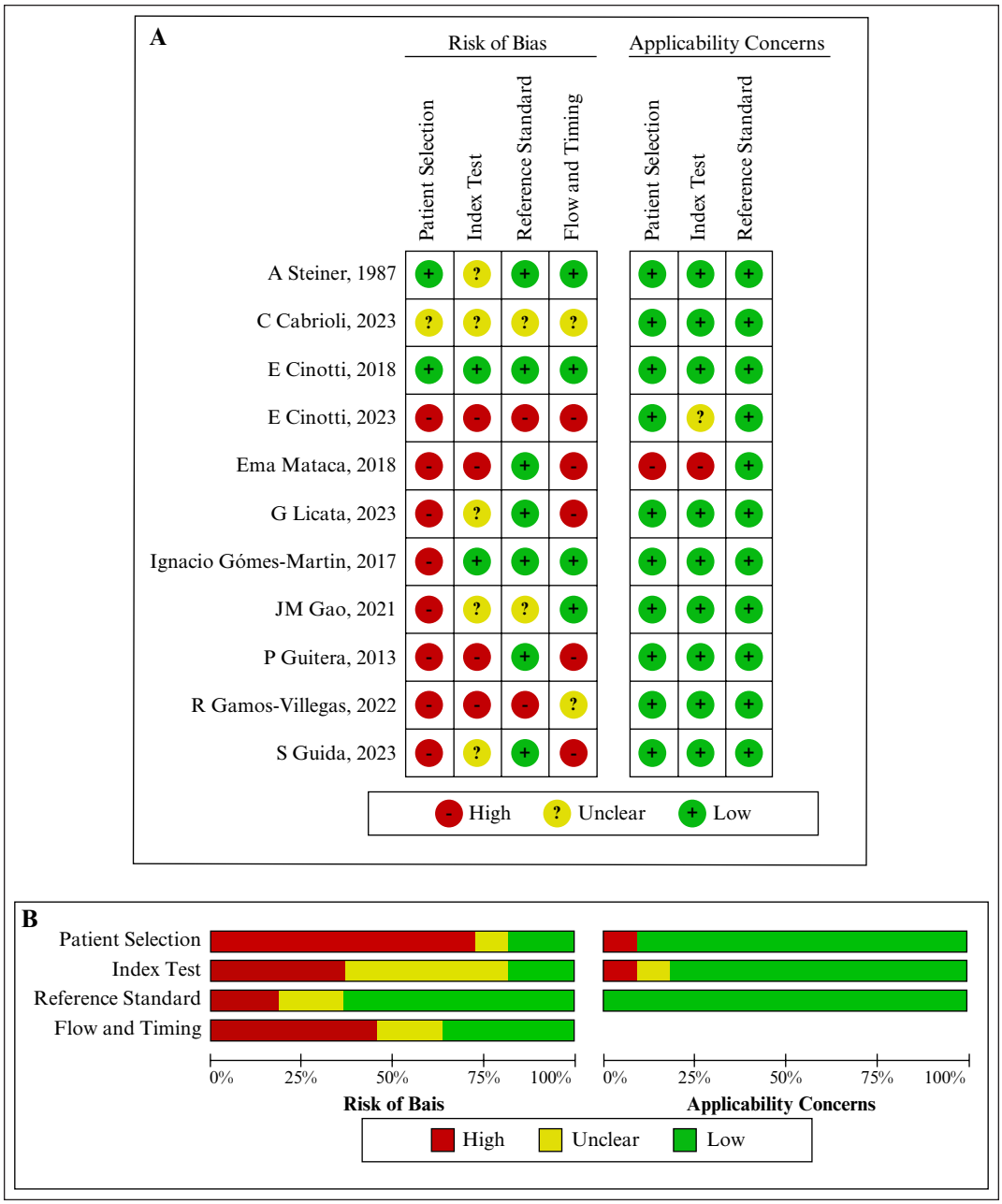


Figure 1. PRISMA flowchart of the selected and analysed studies in this systematic review.

one compared two methods for DMT [28] (*table 1*). A considerable diversity was observed in the experimental design styles of these studies. In addition, the outcomes investigated varied substantially, with studies examining margin mapping and Mohs surgery [18], lesion evolution [21], and presence of subclinical LM [22].

In the study by Steiner *et al.* [17], three clinicians evaluated 318 lesions using both visual inspection and DMT. The investigators were blinded to the results of one of the methods. The lesions analysed included melanoma, dysplastic nevi, junctional nevi, pigmented Spitz nevi, seborrheic keratosis, pigmented basal cell carcinoma, and angiokeratoma. Analysis of the 318 lesions revealed that visual inspection correctly diagnosed 60% of cases. However, DMT correctly diagnosed 84% of the lesions. Specifically, for the diagnosis of LM or LMM, visual inspection correctly identified 67% of the cases, whereas DMT achieved correct diagnoses in 88% of cases. In conclusion, DMT appeared to be useful for the diagnosis of skin lesions compared to naked-eye inspection. Although the authors described that DMT correctly diagnosed 88% of the cases, it is unclear if this value refers to sensitivity of the technique.

Licata *et al.* [19] analysed thin melanomas (<0.5 mm) on photodamaged skin, excluding areas of the face and

Table 1. Description of the 11 studies selected for this systematic review.

Authors	Year	N	Lesions	Methods	Outcomes	DMT Sens.	DMT Spec.	RCM Sens.	RCM Spec.	Blinded evaluators	Direct comparison	Random assignment	Selection of confirmed LM/LMM cases
Steiner et al. [17]	1987	318	LM and LMM	Visual inspection and DMT	Confirmation of malignancy	NR	NR	NR	NR	No	No	No	No (cases selected were suspected for other lesion types)
Guitera et al. [20]	2013	37	LM and LMM	DMT and RCM	Presence of LM or LMM	35%	97%	91%	96%	No	Yes	No	Yes
Gómez-Martín et al. [25]	2017	61	LM	DMT and RCM	Presence of LM or LMM	NR	NR	91.7%	86.8%	Yes	No	No	No (ambiguous pigmented facial macules selected)
Cinotti et al. [24]	2018	201	LM and LMM	DMT and RCM	Confirmation of malignancy	61%	92%	80%	81%	Yes	Yes	No	No (consecutive equivocal lesions)
Mataca, et al. [26]	2018	72	LM and LMM	DMT and RCM	Presence of LM or LMM	NR	NR	NR	NR	Yes	No	No	Yes
Gao et al. [18]	2021	47	LM	Visual inspection, DMT and RCM	Mohs layers required for total excision	NR	NR	NR	NR	No	No	No	Yes
Gamo-Villegas et al. [21]	2022	17	LM and LMM	DMT and RCM	Lesion evolution	NR	NR	NR	NR	No	No	No	Yes
Licata et al. [19]	2023	48	LM	Visual inspection, DMT and RCM	Presence of extrafacial LM	NR	NR	90%	97%	Yes	No	No	Yes
Guida et al. [27]	2023	180	LM and LMM	DMT and RCM	Presence of LM or LMM	NR	NR	NR	NR	Yes	No	No	Yes
Cabrioli et al. [22]	2023	107	LM and LMM	DMT and RCM	Presence of subclinical LM	NR	NR	NR	NR	No	No	Yes	Yes
Cinotti et al. [28]	2023	61	LM and LMM	DMT 20x vs. 400x	Presence of LM or LMM	NR	NR	NR	NR	No	Yes	No	Yes

N: sample size; LM: lentigo maligna; LMM: lentigo maligna melanoma; DMT: dermoscopy; RCM: reflectance confocal microscopy, NR: not reported, Sens: sensitivity, Spec: specificity. Blinded evaluators: with (yes) or without (no) mention of blinded second evaluation; Direct comparison: comparison between DMT and RCM for the correct detection of LM/LMM based on histopathology; Random assignment: the study enrolled a random sample of eligible patients/lesions (or enrolled consecutively) through the clinical setting or database, to reduce selection bias; Selection of confirmed cases: patients were enrolled in the study with (yes) or without (no) prior confirmation of LM/LMM diagnosis. Studies including ambiguous lesions or unconfirmed diagnosis prior to enrollment are also described.

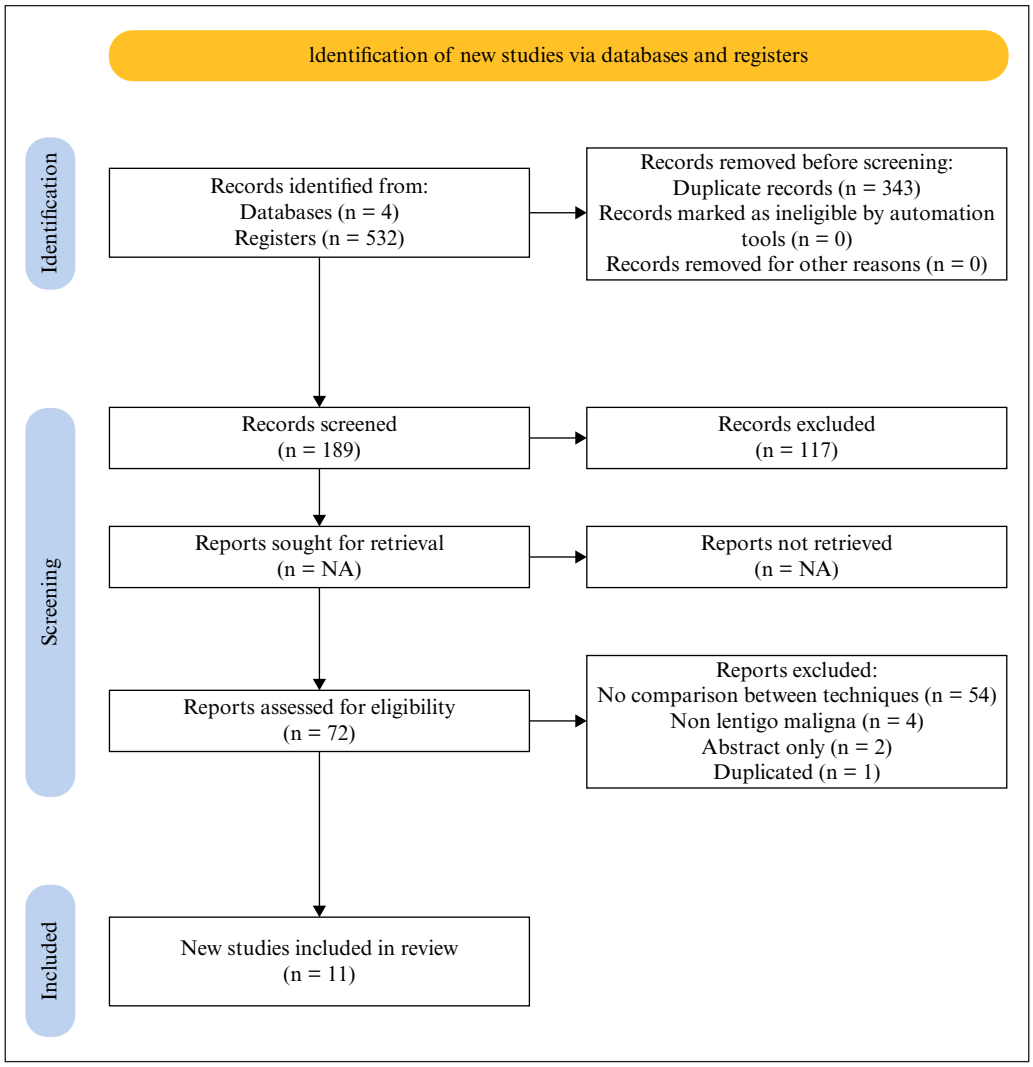


Figure 2. Summary of bias evaluation through the QUADAS-2 algorithm. **A**) Risk of bias analysis of the observational studies included in this systematic review. **B**) Summary of the QUADAS-2 assessment results for the 11 studies included in this review. Figure generated using RevMan software (Cochrane).

neck. A total of 48 extrafacial LM lesions were retrospectively and multicentrically selected. The cases were identified based on findings observed through DMT and visual inspection (epidermal thinning, such as loss of epidermal ridges, prevalence of lentiginous patterns, and actinic damage). The authors also reported sensitivity and specificity scores for melanoma detection with RCM as 90% and 97%, respectively; however, they did not provide the exact numbers of false-positive and false-negative cases and did not directly compare DMT to RCM.

Guitera *et al.* [20] described the use of RCM in the clinical management of challenging LM cases and assessed whether examination findings influenced clinical decisions and outcomes. The study evaluated lesions considered difficult due to either high pigmentation or amelanotic presentation. Thirty-two patients were diagnosed with LM and five with LMM. Each patient underwent at least two biopsies for histopathological confirmation, targeting both positive and negative areas for comparison. Lesions were initially assessed using DMT for margin delineation. Subsequently, margins were recalculated based on information obtained via RCM. Seventeen of the 29 patients with visible lesions showed evidence of subclinical disease extending beyond the 5-mm margin identified by DMT. The visible area was on average 40% smaller than the area treated based on RCM findings. Of the 185 biopsied sites, 60 were histopathologically confirmed as LM. Among these 60 confirmed LM sites, 35% were correctly detected by DMT. However, 65% of the sites were classified as non-malignant, yet histopathology revealed malignancy. In comparison, RCM correctly diagnosed 91% of the lesions as LM, while 9% were misclassified as non-malignant but were histopathologically confirmed as LM. Of the 125 negative biopsy sites, DMT correctly identified 97% as negative. Similarly, RCM accurately classified 96% of the lesions as negative.

Guida *et al.* [27] investigated 180 lesions from 180 patients. The authors emphasized the importance of identifying features such as obliterated follicles and hyperpigmented areas using RCM for the diagnosis and management of LM and LMM. However, comparisons regarding the sensitivity and specificity of the diagnostic methods, as well as agreement between them for the diagnosis of LM and LMM, were not reported.

In the study of Cinotti *et al.* [28], four dermatologists analysed images of patients with pigmented facial lesions that were either scheduled for removal or monitoring after atypical clinical and/or dermatoscopic features had been identified by two dermatologists. A total of 61 skin lesions and 1,138 images were assessed using DMT at 20× and 400× magnification. The use of both 400× and 20× DMT enabled the identification of LM and LMM through the presence of atypical melanocytic proliferation and folliculotropism.

Gamo-Villegas and collaborators [21] analysed 17 patients with LM or LMM lesions confirmed by histopathology, each with at least two follow-up images obtained at three-month intervals. Body mapping was performed using DMT and RCM prior to lesion excision. Through RCM, the authors most frequently observed, in LM, a non-predominant atypical pattern in the epidermis (67%) and a non-predominant atypical pattern at the dermoepidermal junction (DEJ) (100%). In LMM cases, the most frequent findings included a non-predominant atypical pattern in the epidermis (80%), rounded and dendritic cells (40%), and ill-defined margins (40%). The authors did not observe differences between the findings from DMT and RCM, although supporting data for these conclusions were not presented.

Mataca *et al.* [26] evaluated the impact of DMT and RCM on the histopathological diagnosis of LM and LMM. In this study, 72 patients were selected based on confirmed diagnoses of LM or LMM. The authors indirectly compared DMT and RCM, although the primary comparison was between biopsy specimens and complete excisions for the diagnosis of LM and LMM.

In another study, Cabrioli *et al.* [22] evaluated 107 patients diagnosed with LM or LMM by histology (biopsy). Although the authors stated that the selection was randomized, they did not statistically demonstrate that the RCM and DMT groups were equivalent in terms of clinical characteristics. The authors observed that 92% (23/25) of cases had negative margins at first excision. Following the first excision, margins remained positive in approximately 90% of LMM cases using the RCM method; however, around 70% of cases had positive margins detected by DMT. Subsequently, the authors reported that the sensitivity of the RCM technique for detecting subclinical margins was 94.7%, and its specificity was 83.3%. The authors concluded that the RCM method is superior for identifying positive margins.

Cinotti *et al.* [24] analysed 201 patients and 223 lesions including LM, LMM, basal cell carcinomas, and other types of lesions. A consecutive randomized selection was employed, and the groups compared were LM or LMM vs. benign lesion, using visual inspection vs. DMT vs. RCM. For the diagnosis of malignancy, the accuracy assessed by the AUC was 0.86 for DMT and 0.89 for RCM. The sensitivity for malignancy (LM and LMM versus benign lesions) for DMT was 61% and for RCM was 80%. The reported specificity for DMT was 92% and for RCM was 81%. Overall, RCM demonstrated greater sensitivity compared to DMT, while DMT had higher specificity in the diagnosis of LM and LMM. The authors state that the diagnosis of malignant lesions benefits from the combined use of RCM and DMT to achieve greater accuracy.

Lastly, in the study of Gómez-Martín *et al.* [25], three evaluators analysed lesions from 61 White patients, including 63 ambiguous lesions and 12 controls (photodamaged regions). The authors identified significant correlations between rhomboidal structures observed via DMT and large pagetoid cells identified by RCM (κ=0.44), as well as with the presence of three or more atypical cells at the DEJ (κ=0.32). Furthermore, pigmented follicular openings identified through DMT were significantly correlated with the location of large pagetoid cells and with three (κ=0.49) or more atypical cells at the DEJ (κ=0.37). Lastly, granular dust detected by DMT correlated with large pagetoid cells observed via RCM (κ=0.31). No sensitivity and specificity scores for DMT and RCM were presented in the study.

Discussion

Summary of evidence

The objective of this systematic review was to investigate the literature comparing DMT and RCM techniques in the diagnosis of LM and LMM confirmed by histopathology, considering their accuracy, sensitivity, and specificity. After retrieving 532 studies through search platforms, 11 were selected for qualitative analysis of findings. The analysis of these publications revealed that there is still a significant gap in the literature regarding methods for diagnosis of LM and LMM. The lack of consistency and clarity in study designs, statistical analysis, and data reporting strongly impair any conclusions that could be drawn in this review.

The risk of bias assessment using the QUADAS-2 tool revealed significant limitations across the 11 studies evaluated. The majority of the publications included in this review presented a high risk of bias in patient selection, as participants had already been previously diagnosed with LM or LMM when selected for the study [24, 25], which favours lesions to be more easily recognized. Selection bias due to disease recognition was mitigated in some studies that reported blinding of investigators to patient diagnoses. Additionally, only one study was classified as having a low risk of bias across all domains assessed by the QUADAS-2 tool [24]. Also, only two studies clearly stated prospective designs, in which ambiguous skin lesions were analysed without prior diagnosis confirmation of LM and LMM [25, 29].

Non-random participant recruitment may lead to data collection and conclusions that do not reflect the general population. Given the importance of diagnosis via DMT or RCM, studies should be conducted using random population sampling, including patients with histopathologically confirmed LM or LMM and others without such confirmation. This approach would allow for

clearer identification of the sensitivity and specificity of non-invasive diagnostic methods. A Cochrane systematic review aimed to determine the diagnostic accuracy of basal cell carcinoma using visual inspection and DMT alone or in combination [30]. The authors also applied the QUADAS-2 tool to assess the risk of bias in the included studies. Similar to our findings, the Cochrane review authors identified a high risk of participant selection bias. In most of the studies analysed, participant selection was based on biopsy or clinical inspection confirmation. Consequently, the lesions assessed via visual inspection or DMT by investigators were, in most cases, already confirmed as basal cell carcinoma. Selection bias significantly impacts diagnostic research outcomes [31] and, therefore, deserves greater attention in study design.

Another critical issue identified was the lack of randomization in patient selection across many studies, which may have introduced additional bias. For example, in the study of Cabrioli *et al.* [22], although the authors claimed random selection, there was no statistical evidence confirming equivalence between the compared groups (RCM versus DMT), making interpretation of the results more challenging. The absence of randomization in diagnostic studies raises significant concerns regarding the validity and applicability of their findings. Although randomized clinical trials (RCTs) are considered the gold standard for evaluating treatment efficacy, the application of randomization in studies assessing diagnostic tests remains limited [32]. This limitation may introduce biases that affect the interpretation of diagnostic accuracy and clinical relevance. In their study, Rodger *et al.* [32] emphasize the importance of conducting RCTs for disease diagnosis, as they help eliminate or minimize many biases that compromise the internal validity of diagnostic accuracy. Through randomization, these RCTs ensure that context-related bias is eliminated, since both competing strategies are evaluated in groups with a similar disease prevalence. In the case of this systematic review, only one study reported that samples were obtained through randomization. This finding highlights the urgent need for the conduct of randomized diagnostic studies.

The absence of direct comparisons between DMT and RCM was one of the most frequently observed limitations among the evaluated studies. Many investigations examined the techniques either in isolation or indirectly, without providing robust comparative sensitivity and specificity data [26, 27]. For instance, Guida *et al.* [27] reported significant associations between DMT and RCM findings, such as pigment patterns and subcellular structures, but did not present quantitative diagnostic performance data that would allow for an in-depth comparative analysis.

A review of systematic reviews and meta-analyses conducted by Takwoingi *et al.* [33] aimed to investigate the availability of direct comparative studies on diagnostic accuracy and to assess whether summary estimates of accuracy differ between meta-analyses of non-comparative and comparative studies. Of the 6,915 publications analysed across 248 reviews, only 2,113 (31%) were comparative in nature. The authors also observed that in 10 meta-analyses, non-comparative studies ranked diagnostic tests in the opposite direction compared to comparative studies. Moreover, the differences in accuracy estimates between non-comparative and comparative studies were statistically significant. The study concluded that evidence derived from non-comparative studies often diverges from that obtained through comparative studies. Furthermore, given that non-comparative studies are frequently used as evidence in diagnostics, clinicians and public health programs must re-evaluate their practices when relying on findings that may, in many cases, contradict the actual scenario.

The lack of application of appropriate statistical methods to calculate fundamental metrics such as sensitivity, specificity, and AUC was a recurring limitation among the 11 selected studies. Only a few studies reported such metrics, making it difficult to interpret the diagnostic performance of visual inspection, DMT, and RCM individually. For example, although Licata *et al.* [19] reported a sensitivity of 90% and a specificity of 97% for RCM, data for DMT were not presented, preventing direct comparisons between the two non-invasive diagnostic methods.

When evaluating diagnostic tests, it is essential to consider both their ability to confirm the presence of disease and to exclude disease in healthy individuals [34]. When the result of a diagnostic test is binary (positive or negative), the traditional approach to evaluating its accuracy relies on calculating sensitivity and specificity in comparison to a gold standard. Therefore, to determine the diagnostic capability of an index test, it must be directly compared with the gold standard. However, few studies selected for this systematic review reported sensitivity and specificity results for comparative diagnostic methods. These studies prioritized the description of dermatological findings observed through different techniques; however, using such findings to establish diagnoses without statistically demonstrating their ability to distinguish malignant from benign tumours reduces the clinical applicability of the results. The outcomes analysed in the included studies exhibited substantial heterogeneity, making it difficult to compare results across studies. While some investigations focused exclusively on the diagnosis of LM and LMM [24, 27], others explored more specific endpoints such as surgical margin delineation [18, 22] or post-treatment recurrence [20]. This variability reflects the broad range of potential applications for these techniques, but also highlights the need for standardization in outcome criteria.

For example, the study by Guitera *et al.* [20] highlighted the clinical impact of RCM in reducing recurrences, whereas Licata *et al.* [19] focused on the diagnosis of thin melanomas in extrafacial areas. Although both studies provided relevant findings, the low uniformity of investigated outcomes limits direct comparisons between studies. A study examining outcome heterogeneity in renal cancer research demonstrated that the definition, selection, and reporting of outcomes in studies hampers evidence-based clinical practice [35]. Future studies should consider standardizing outcomes, especially those related to diagnostic accuracy and clinical impact, to facilitate comparison and strengthen the robustness of the evidence base.

Based on the findings of the studies and the risk of bias analysis, it is currently challenging to prioritize either

the DMT or RCM technique for the diagnosis of LM and LMM. The available evidence is limited, the results present significant constraints, and the substantial variability in experimental designs, statistical analyses, and outcomes prevents any definitive decision-making. Additionally, the non-random and non-consecutive selection of patients and the use of diagnostic confirmation as a criterion for patient inclusion in the analysed studies further challenges us to define which non-invasive method is more appropriate for diagnosing LM and LMM.

A 2024 study by Dradi *et al.* [36] investigated the relevance of pigmented parallel lines in the diagnosis of LM, utilizing and comparing DMT and RCM in 22 cases of LM and LMM. However, as with the other studies included in this review, patients were enrolled after having been histologically diagnosed with LM or LMM in a non-consecutive setting. Moreover, the study did not perform direct comparisons between the methods and did not present statistical results pertaining to such comparisons. Therefore, the urgent need is underscored for diagnostic studies with: (1) larger sample sizes; (2) randomly collected samples prior to histopathological diagnosis confirmation; and (3) appropriate strategies for the comparison of diagnostic methods.

Limitations

This systematic review presents several limitations that must be acknowledged. Most notably, we did not perform a meta-analysis with the data obtained from the selected studies. The reasons are the heterogeneity of data obtained, outcomes and lack of direct comparisons between the DMT, RCM and visual inspection methods for LM and LMM diagnosis. Moreover, as discussed, the methodological and reporting limitations restrict our ability to draw definitive conclusions about the comparative diagnostic value of visual inspection, DMT, and RCM for LM and LMM.

Conclusions

This systematic review highlights the lack of consistency across studies investigating the ability of RCM versus DMT to correctly diagnose LM and LMM. Strong methodological limitations, such as selection bias, lack of randomization, absence of direct comparisons between techniques and selection of cases already confirmed as LM and LMM undermine the reliability of results from these studies. These findings show that current evidence for which technique should be used for LM and LMM diagnosis remains uncertain. Future studies should involve larger, randomized samples, prospective designs, and standardized comparative analyses to enhance the applicability of the findings and conduct of other systematic reviews and meta-analyses. ■

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